

MODULE 6

1. Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address.

The ARP request is a broadcast frame sent to all devices in the network.

- Destination MAC: FF:FF:FF:FF:FF:FF
- Sender IP: Host IP address
- Sender MAC: Host MAC address
- Target IP: IP address being resolved
- Target MAC: Unknown (00:00:00:00:00:00)

The ARP reply is a unicast frame sent back to the requester.

- Destination MAC: Requester's MAC
- Sender IP: Target IP
- Sender MAC: Target device MAC
- Target IP: Requester IP
- Target MAC: Requester MAC

Role of Sender IP and MAC

- The Sender IP address provides logical identification and ensures the reply is sent to the correct IP host.
- The Sender MAC address provides physical identification required for frame delivery in the local network.

30766 -482.7119944...	Intel_e3:e5:09	Broadcast	ARP	42 Who has 10.13.242.36? Tell 10.13.242.153
30767 -482.7004118...	1a:37:0b:1e:6a:cd	Intel_e3:e5:09	ARP	42 10.13.242.36 is at 1a:37:0b:1e:6a:cd

ARP cache on the laptop

```
Select Administrator: Windows PowerShell

Install the latest PowerShell for new features and improvements! https://aka.ms/WindowsPowerShellLatestVersion

PS C:\WINDOWS\system32> arp -a

Interface: 10.13.242.153 --- 0xc
    Internet Address      Physical Address      Type
    10.13.242.36          1a-37-0b-1e-6a-cd    dynamic
    10.13.242.255         ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0x11
    Internet Address      Physical Address      Type
    192.168.56.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

PS C:\WINDOWS\system32> arp -d *
PS C:\WINDOWS\system32> arp -a

Interface: 10.13.242.153 --- 0xc
    Internet Address      Physical Address      Type
    224.0.0.22            01-00-5e-00-00-16    static

Interface: 192.168.56.1 --- 0x11
    Internet Address      Physical Address      Type
    224.0.0.22            01-00-5e-00-00-16    static

PS C:\WINDOWS\system32> ping 10.13.242.36

Pinging 10.13.242.36 with 32 bytes of data:
Reply from 10.13.242.36: bytes=32 time=9ms TTL=64
Reply from 10.13.242.36: bytes=32 time=10ms TTL=64
Reply from 10.13.242.36: bytes=32 time=22ms TTL=64
Reply from 10.13.242.36: bytes=32 time=2ms TTL=64

Ping statistics for 10.13.242.36:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 22ms, Average = 10ms
PS C:\WINDOWS\system32>
```

ARP Request

```
Wireshark - Packet 30766 - WiFi

Frame 30766: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
Ethernet II, Src: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... 01. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... 01. .... = IG bit: Group address (multicast/broadcast)
  Source: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
  Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    Sender IP address: 10.13.242.153 (10.13.242.153)
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.13.242.36 (10.13.242.36)

0000  ff ff ff ff ff ff b0 3c dc e3 e5 09 08 06 00 01  ....-<-j-----
0010  08 00 06 04 00 01 b0 3c dc e3 e5 09 0a 0d f2 99  ....<-j-----
0020  00 00 00 00 00 00 0a 0d f2 24  ....-j-----
```

ARP Response

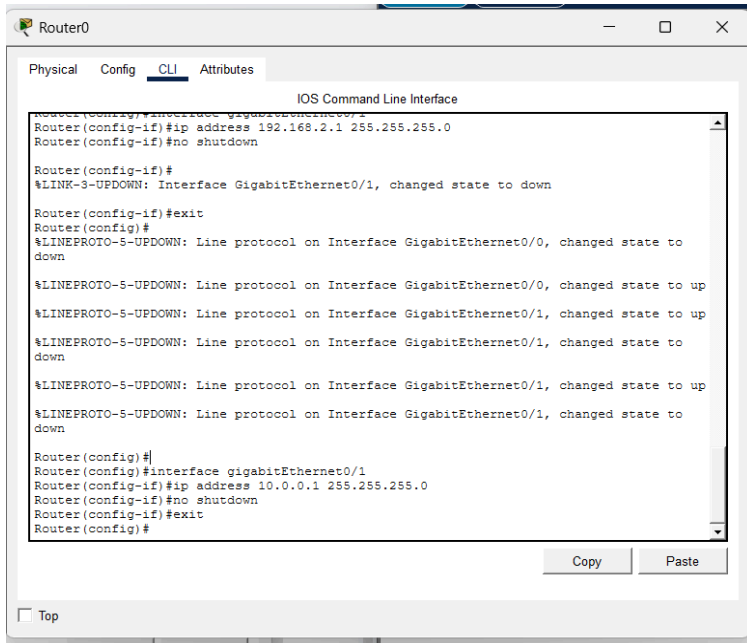
```
Wireshark - Packet 30767 - WiFi

Frame 30767: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{8D44E5C3-21A6-415F-A343-6B9E6FBC225F}, id 0
Ethernet II, Src: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd), Dst: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
  Destination: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    .... 00. .... = LG bit: Globally unique address (factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  Source: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd)
    .... 01. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... 00. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 0]
  Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: 1a:37:0b:1e:6a:cd (1a:37:0b:1e:6a:cd)
    Sender IP address: 10.13.242.36 (10.13.242.36)
    Target MAC address: Intel_e3:e5:09 (b0:3c:dc:e3:e5:09)
    Target IP address: 10.13.242.153 (10.13.242.153)

0000  b0 3c dc e3 e5 09 1a 37 0b 1e 6a cd 08 06 00 01  ....7-j-----
0010  08 00 06 04 00 02 1a 37 0b 1e 6a cd 0a 0d f2 24  ....7-j-----
0020  b0 3c dc e3 e5 09 0a 0d f2 99  ....<-j-----
```

2. **Manually configure static routes on a router to direct packets to different subnets. Use the ip route command and verify connectivity using ping and traceroute.**

Configuring the interfaces on router0



The screenshot shows the CLI window for Router0. The configuration commands entered are:

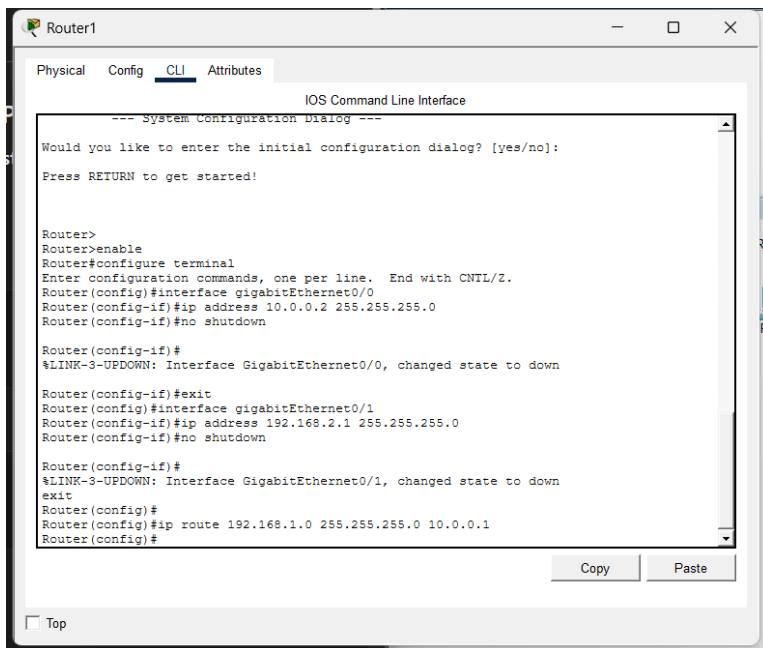
```
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down

Router(config-if)#exit
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

Router(config)#
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
```

Configuring the interfaces on router1



The screenshot shows the CLI window for Router1. The configuration commands entered are:

```
--- System Configuration Dialog ---

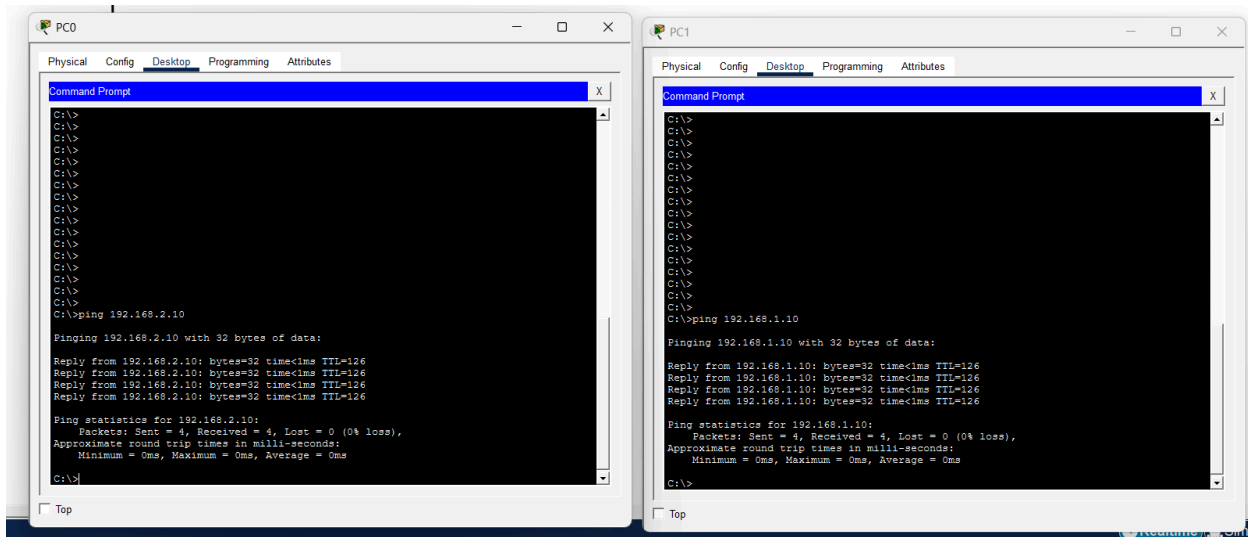
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down

Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
exit
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
```



- Static routing was configured in Cisco Packet Tracer to enable communication between different subnets.
- The network topology consists of routers and PCs.
- IP addresses were then assigned to the router interfaces, where GigabitEthernet0/0 was configured with 192.168.1.1/24 and GigabitEthernet0/1 with 192.168.2.1/24, followed by enabling the interfaces using the no shutdown command.
- Next, IP addresses were assigned to the end devices: PC1 was configured with IP address 192.168.1.10, subnet mask 255.255.255.0, and default gateway 192.168.1.1, while PC2 was configured with IP address 192.168.2.10, subnet mask 255.255.255.0, and default gateway 192.168.2.1.
- Since multiple routers were used, static routes were manually configured using the ip route command, where routes to remote networks were defined via the next-hop IP addresses (10.0.0.2 and 10.0.0.1). Finally, connectivity between the networks was verified using the ping command and the traceroute command.

- 3. Given a network address of 10.0.0.0/24, divide it into 4 equal subnets.
Calculate the new subnet mask.
Determine the valid host range for each subnet.
Assign IP addresses to devices in Packet Tracer and verify connectivity.**

Network: 10.0.0.0/24

4 equal subnets

- Original mask: /24
- Need 4 subnets

New Subnet Mask = 255.255.255.192

Each subnet size: 64

Subnet 1

- Network: 10.0.0.0
- Range: 10.0.0.1 – 10.0.0.62
- Broadcast: 10.0.0.63

Subnet 2

- Network: 10.0.0.64
- Range: 10.0.0.65 – 10.0.0.126
- Broadcast: 10.0.0.127

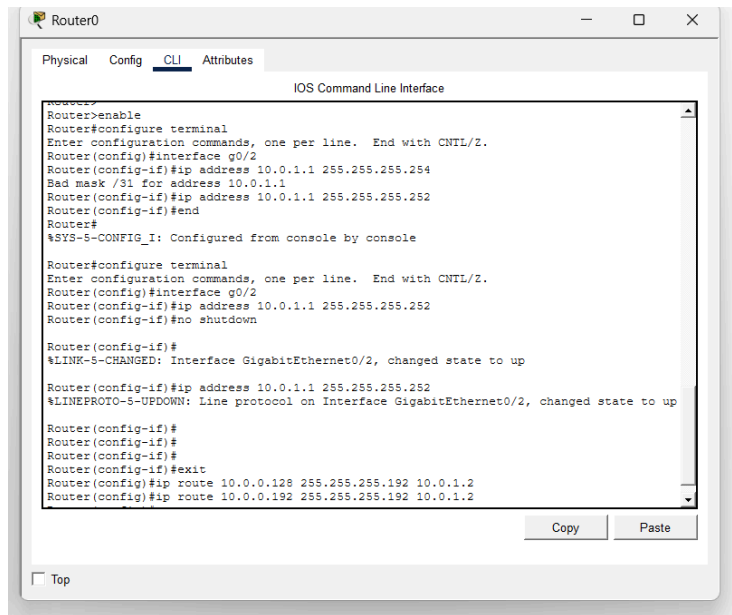
Subnet 3

- Network: 10.0.0.128
- Range: 10.0.0.129 – 10.0.0.190
- Broadcast: 10.0.0.191

Subnet 4

- Network: 10.0.0.192
- Range: 10.0.0.193 – 10.0.0.254
- Broadcast: 10.0.0.255

Configuring the routes and interfaces



```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

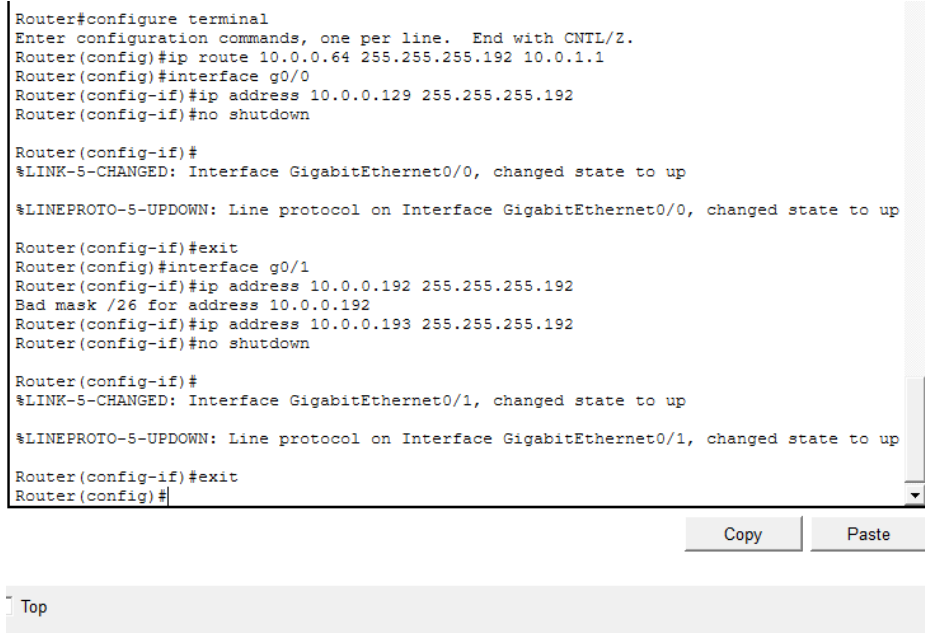
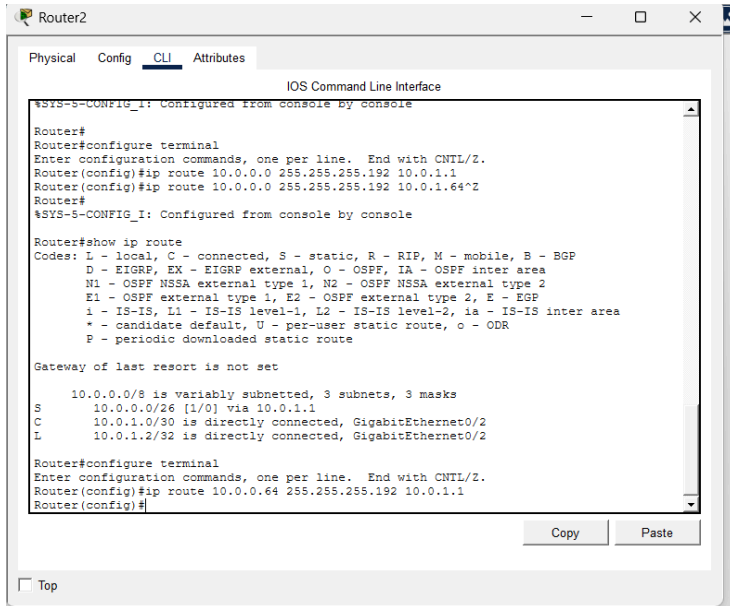
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.0.65 255.255.255.192
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
Router(config)#
```

Copy

Paste



IP addresses of the pcs in different subnets

PC0

Physical Config **Desktop** Programming Attributes

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.10

Subnet Mask 255.255.255.192

Default Gateway 10.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BCFF:FE1E:EE82

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.0.0.70

Subnet Mask 255.255.255.192

Default Gateway 10.0.0.65

DNS Server 0.0.0.0

IPv6 Configuration

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

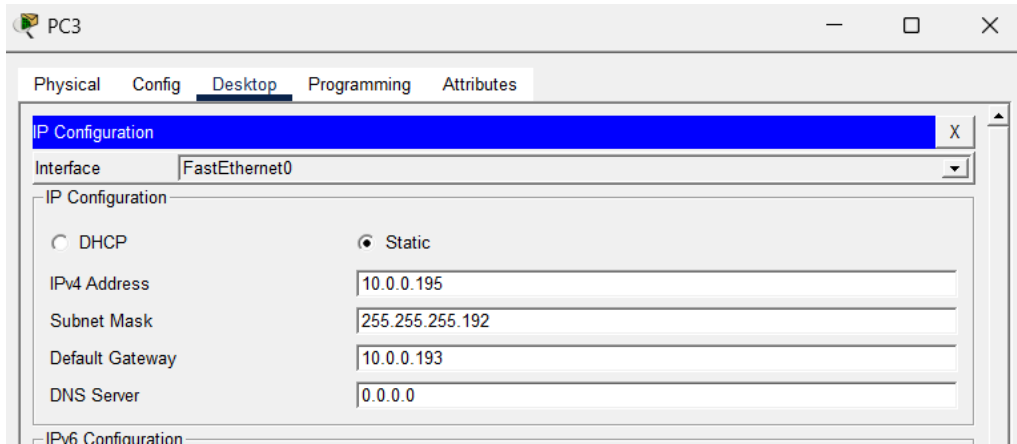
IPv4 Address 10.0.0.140

Subnet Mask 255.255.255.192

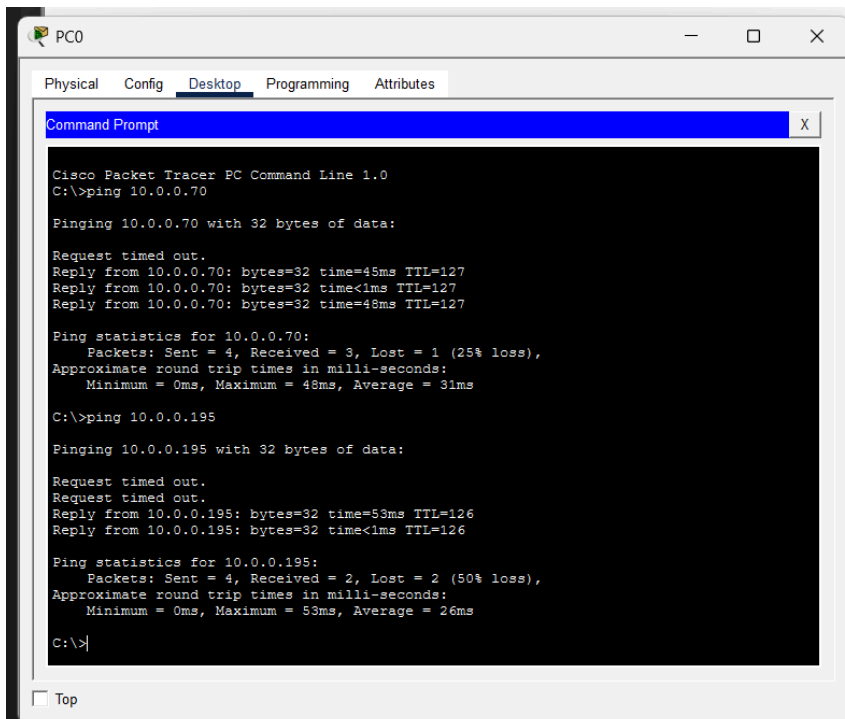
Default Gateway 10.0.0.129

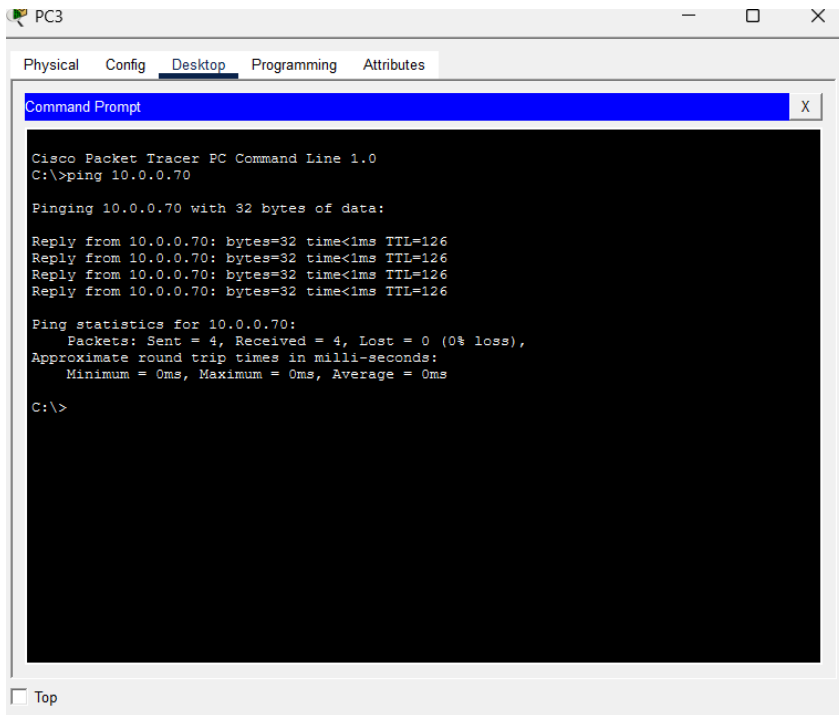
DNS Server 0.0.0.0

IPv6 Configuration



Pinging to verify the connectivity





4. You are given three IP addresses: 192.168.10.5, 172.20.15.1, and 8.8.8.8.
Identify the class of each IP address.
Determine if it is private or public.
Explain how NAT would handle a private IP when accessing the internet.

1. 192.168.10.5
Class: C
Type: Private

2. 172.20.15.1
Class: B
Type: Private

3. 8.8.8.8
Class: A
Type: Public

Private and Public IPs

- Private IP addresses are reserved for use within local networks and are not routable on the internet.
- Public IP addresses are globally unique and used for communication over the internet.

Private IP ranges include:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

NAT:

Private IP addresses require NAT to communicate with the internet, while public IP addresses can communicate directly. NAT enables multiple devices in a local network to share a single public IP address efficiently.

When a device with a private IP address (e.g., 192.168.10.5) wants to access the internet, it cannot do so directly because private IPs are not routable.

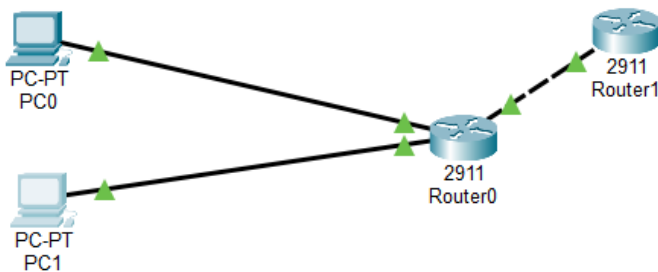
NAT is performed by a router to solve this:

1. The device sends a request to an external server (e.g., 8.8.8.8).
2. The router replaces the private IP address with its own public IP address.
3. It also assigns a unique port number to track the session.
4. The request is sent to the internet using the public IP.
5. When the response comes back, the router uses the port number to map it back to the original private IP device.

5. In Cisco Packet Tracer, configure NAT on a router to allow internal devices (192.168.1.x) to access the internet.

Test connectivity by pinging an external public IP.

Capture the traffic in Wireshark and analyze the source IP before and after NAT translation.

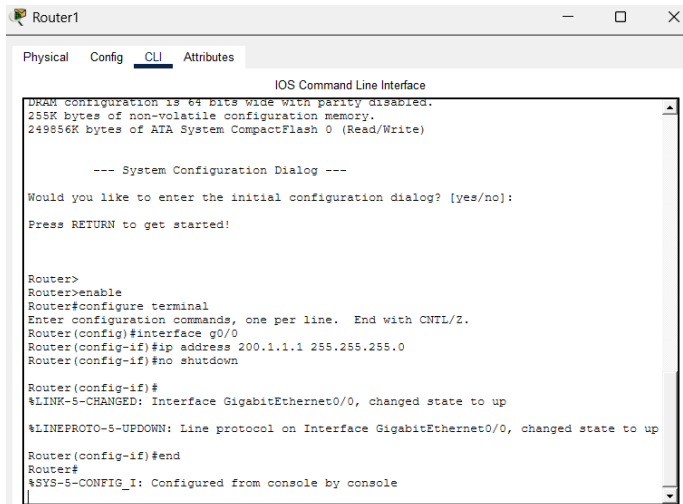


Router 0

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
down
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 200.1.1.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
Router>
Router>
Router>enable
Router#configure
Configuring from terminal, memory, or network [terminal]? terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 1



The screenshot shows the Router1 CLI window with the 'CLI' tab selected. The window title is 'Router1'. The interface displays the following text:

```
IOS Command Line Interface

DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

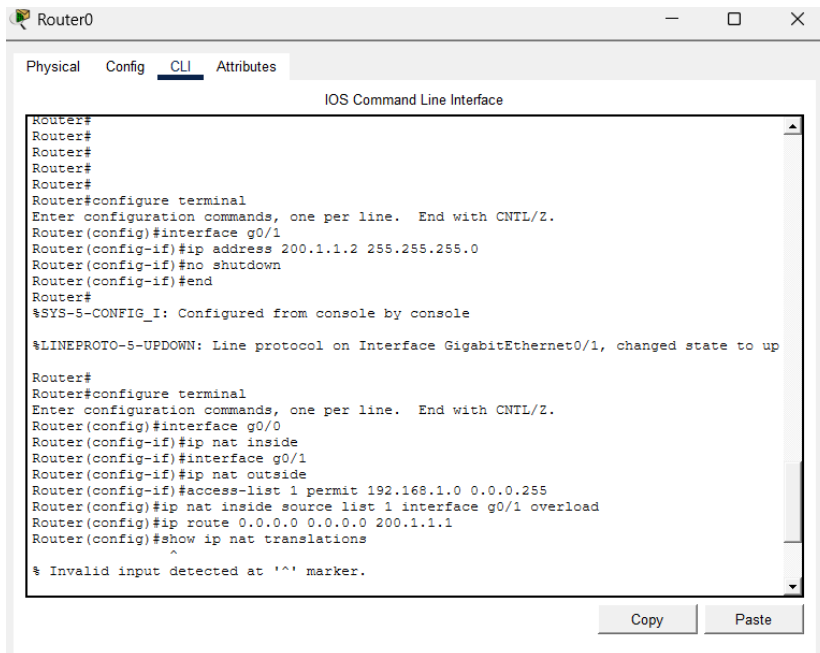
Press RETURN to get started!

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

NAT configuration



The screenshot shows the Router0 CLI window with the 'CLI' tab selected. The window title is 'Router0'. The interface displays the following text:

```
IOS Command Line Interface

Router#
Router#
Router#
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 200.1.1.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#ip route 0.0.0.0 0.0.0.0 200.1.1.1
Router(config)#show ip nat translations
^
% Invalid input detected at '^' marker.
```

At the bottom of the window, there are 'Copy' and 'Paste' buttons.

Nat translation

```
Router#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 200.1.1.2:1       192.168.1.10:1    200.1.1.1:1        200.1.1.1:1
icmp 200.1.1.2:2       192.168.1.10:2    200.1.1.1:2        200.1.1.1:2
icmp 200.1.1.2:3       192.168.1.10:3    200.1.1.1:3        200.1.1.1:3
icmp 200.1.1.2:4       192.168.1.10:4    200.1.1.1:4        200.1.1.1:4

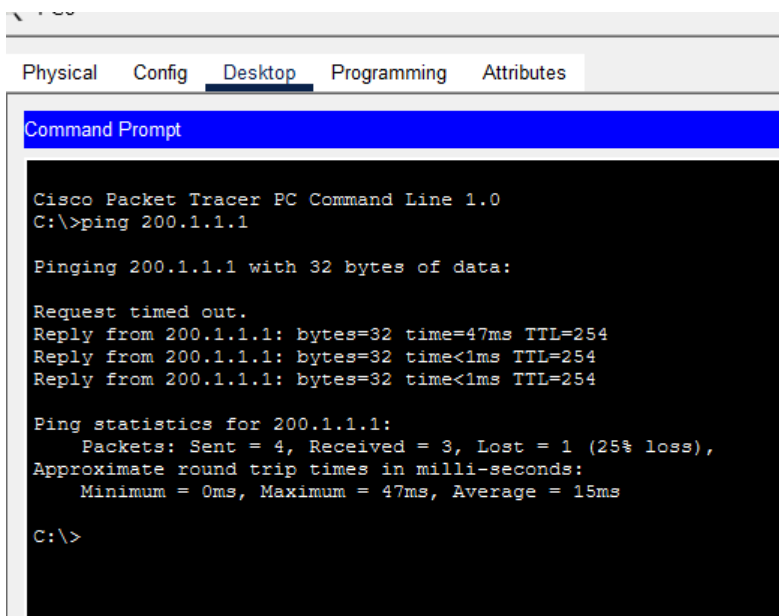
Router#
```

Pinging to verify connectivity

```
Pinging 200.1.1.2 with 32 bytes of data:

Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



The PCs were assigned private IP addresses (192.168.1.10 and 192.168.1.20) with a subnet mask of 255.255.255.0 and a default gateway of 192.168.1.1.

The router connected to the internal network was configured with IP address 192.168.1.1 on the inside interface and 200.1.1.2 on the outside interface.

Another router was configured to simulate the internet with IP address 200.1.1.1. Network Address Translation (NAT) was configured on the internal router by designating inside and outside interfaces, creating an access control list to permit the private network, and enabling NAT overload using the command "ip nat inside source list 1 interface g0/1 overload." A default route was also configured to forward traffic to the external network.

The NAT configuration was successfully verified by pinging the external IP address from the PCs and observing address translation using the "show ip nat translations" command.